

Operating Systems and Linux Administration

Study Notes for Exit Exam Preparation

This short material is designed for testing AI summary, RAG question answering, flashcards, quiz generation, and audio summary features in UniPrep AI.

Learning Outcomes

- Explain the purpose and responsibilities of an operating system.
- Describe process management, CPU scheduling, memory management, and file systems.
- Distinguish between Linux root users, human users, and service accounts.
- Interpret Linux file permissions and modify them using `chmod`, `chown`, and `chgrp`.
- Identify key security practices used in Linux system administration.

1. What Is an Operating System?

An operating system is system software that manages computer hardware and provides services for application programs. It acts as an interface between users, applications, and hardware resources such as the processor, memory, storage devices, and input/output devices.

The main goals of an operating system are convenience, efficiency, and reliability. Convenience means making the computer easier to use. Efficiency means using resources such as CPU time and memory effectively. Reliability means keeping the system stable, secure, and available even when multiple programs are running.

Core operating system responsibilities include process management, memory management, file system management, device management, user management, and security enforcement.

2. Process Management

A process is a program in execution. When a user opens a browser, runs a Python script, or starts a database server, the operating system creates one or more processes. Each process has its own process identifier, memory area, program counter, and execution state.

Common process states include new, ready, running, waiting, and terminated. A process is ready when it can run but is waiting for CPU time. It is running when the CPU is executing its instructions. It is waiting when it needs input/output or another event before it can continue.

CPU scheduling decides which ready process should run next. Examples of scheduling algorithms include First Come First Served, Shortest Job First, Round Robin, and Priority Scheduling. Round Robin is common in time-sharing systems because each process receives a small time slice.

3. Memory Management

Memory management controls how main memory is allocated and protected. Each process needs memory for its code, data, stack, and heap. The operating system must ensure that one process cannot accidentally or maliciously overwrite the memory of another process.

Virtual memory allows the system to use disk storage as an extension of RAM. This makes programs believe they have a large continuous address space even when physical memory is limited. Paging divides memory into fixed-size pages and frames, making allocation more flexible and reducing external fragmentation.

Thrashing occurs when the system spends more time swapping pages between memory and disk than executing useful instructions. It usually happens when too many processes compete for limited memory.

4. File Systems and Storage

A file system organizes data on storage devices. It defines how files are named, stored, retrieved, protected, and organized into directories. Linux uses a single-rooted hierarchy that begins at the root directory represented by a forward slash (/).

Important Linux directories include /home for user files, /etc for configuration files, /var for variable data such as logs, /bin and /usr/bin for commands, /tmp for temporary files, and /root for the root user home directory.

Mounting attaches a file system to a directory in the existing hierarchy. For example, a USB drive may be mounted under /media or /mnt. The mount command attaches file systems, while umount safely detaches them.

Directory	Common Purpose
/home	Personal files of regular users
/etc	System and application configuration files
/var	Logs, cache, mail, and variable data
/tmp	Temporary files that may be removed automatically
/bin and /usr/bin	Essential and user-level commands

5. Linux User Accounts

Linux is a multi-user operating system. Multiple users can have accounts on the same machine, and the system controls what each user is allowed to do. The root account is the superuser account and has full administrative privileges. Because root can modify system files and install software, it should be used carefully.

Human user accounts are created for real people. Service accounts are created for software services such as web servers, databases, or background daemons. Service accounts improve security because each service can run with only the permissions it needs.

User information is commonly stored in /etc/passwd, while password hashes and password aging information are stored in /etc/shadow. The id command shows a user ID, group ID, and group memberships. The useradd, usermod, and userdel commands create, modify, and remove users.

6. Linux Groups

Groups are used to assign permissions to multiple users at once. A user can belong to a primary group and several supplementary groups. The groupadd command creates a group, groupmod modifies a group, and groupdel removes a group. Adding a user to a group is common when granting shared access to project files.

Good administration avoids giving every user administrator-level permissions. Instead, the principle of least privilege should be followed. This means users and services receive only the permissions needed to perform their tasks.

7. File Permissions

Linux file permissions control who can read, write, or execute files and directories. Every file has an owner, a group, and permission bits for user, group, and others. The permission letters are r for read, w for write, and x for execute.

For files, read means viewing file contents, write means modifying contents, and execute means running the file as a program or script. For directories, read means listing names, write means creating or deleting entries, and execute means entering the directory or accessing items inside it.

Permissions can be written symbolically, such as `chmod u+x script.sh`, or numerically, such as `chmod 755 script.sh`. In numeric form, read is 4, write is 2, and execute is 1. Therefore, 7 means read + write + execute, 5 means read + execute, and 0 means no permission.

Command	Purpose
<code>chmod</code>	Change file or directory permissions
<code>chown</code>	Change owner of a file or directory
<code>chgrp</code>	Change group ownership
<code>ls -l</code>	Display long listing with permissions and ownership
<code>id</code>	Show user and group identity information

8. Access Control Models

Discretionary Access Control, or DAC, allows resource owners to decide who can access their files. Traditional Linux file permissions are an example of DAC. Role-Based Access Control, or RBAC, grants permissions based on job roles instead of individual users. Mandatory Access Control, or MAC, enforces strict policies defined by the system or organization, and users cannot freely change those policies.

DAC is flexible but can be risky if owners assign permissions carelessly. RBAC is useful in organizations because it maps permissions to responsibilities. MAC provides stronger control for high-security environments, but it is more complex to manage.

9. System Administration Responsibilities

System administrators are responsible for keeping systems secure, reliable, and available. Their work includes installing software, managing users and groups, configuring services, monitoring system performance, applying updates, checking logs, planning backups, and responding to incidents.

Network administrators focus on connectivity and secure communication. Their tasks may include IP addressing, DNS configuration, firewall rules, routing, VPN setup, and monitoring network availability. In small organizations, one administrator may perform both system and network administration roles.

10. Security Practices

Security is a continuous process. Common Linux security practices include using strong passwords, disabling unnecessary services, applying updates, using firewalls, reviewing logs, limiting root access, using sudo instead of direct root login, and setting correct file permissions.

Authentication verifies who a user is. Authorization determines what the user is allowed to do. Auditing records actions so administrators can review what happened. These three ideas work together to protect systems and investigate problems.

Backups are also part of security and reliability. A system may be secure against attackers but still fail because of hardware damage, accidental deletion, or software errors. Good backups should be regular, tested, and stored safely.

11. Quick Exam Review

- An operating system manages hardware resources and provides services to applications.
- A process is a program in execution and can move through states such as ready, running, waiting, and terminated.
- Round Robin scheduling gives each process a time slice and is useful for time-sharing systems.
- Virtual memory uses disk storage to extend the apparent size of RAM.
- Linux file systems use a single-rooted hierarchy beginning at /.
- The root account has full administrative privileges and should be used carefully.
- chmod changes permissions, chown changes file owner, and chgrp changes group ownership.
- The principle of least privilege means giving users and services only the access they need.
- DAC is owner-controlled, RBAC is role-based, and MAC is policy-enforced.
- Security includes authentication, authorization, auditing, updates, firewalls, and backups.

Suggested Practice Questions

Use this material to ask UniPrep AI questions such as: What are the responsibilities of an operating system? What is the difference between DAC, RBAC, and MAC? How do chmod, chown, and chgrp differ? Why is least privilege important in Linux administration?